

Modelling & Learning

ECMM422

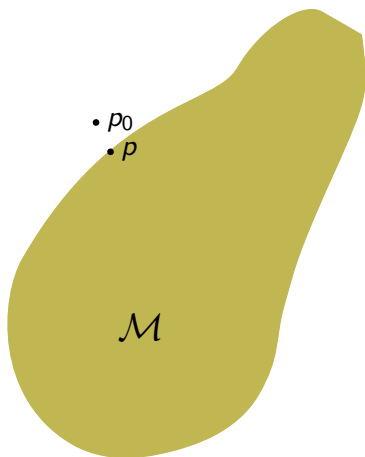
Fabrizio Costa

Winter 2026



Outline

- 1 Modelling
 - Unsupervised learning
 - Supervised learning
- 2 Error functions
- 3 Maximum Likelihood
- 4 Maximum Likelihood in unsupervised learning
 - Learning the mean
 - Kullback-Leibler divergence
- 5 Maximum Likelihood in supervised learning
 - Regression
 - Classification



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Modelling: Unsupervised learning

Given: data

$$\mathcal{D} = \{x_n\}_{n=1}^N$$

and a *parametrised generative model* describing how the data might be generated, $p(x; w)$, depending on parameters w .

Goal is to learn the set of parameters that best describe the data.

Parametrised models

Gaussian distribution:

$$p(x, w) = \mathcal{N}(x; \mu, \sigma^2)$$

Mixture model:

$$p(x, w) = \sum_{m=1}^M \omega_m \mathcal{N}(x; \mu_m, \sigma_m^2)$$

Modelling: Supervised learning

Given data: $\mathcal{D} = \{(x_n, t_n)\}_{n=1}^N$

x patterns: inputs,
independent variables
 t targets: outputs,
dependent variables

Model data as being generated by:

$$t = \mathcal{F}(x; w)$$

w model parameters

Systematic component, which specifies the average response.
Deterministic model of the system.

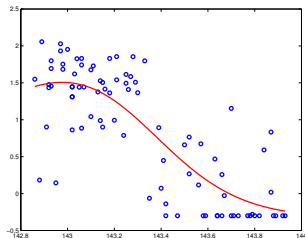
Random component, which quantifies the variability observed at from observation to observation.

Regression error function

Model data generation as sum of deterministic function and Gaussian noise:

$$t_n = \mathcal{F}(x_n, w) + \epsilon_n \quad \epsilon_n \sim \mathcal{N}(0, \sigma^2)$$

$$t_n = y_n + \epsilon_n$$



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Quick recap on probabilities definitions

Let X, Y be two random variables

- ▶ **sum rule:** $p(X) = \sum_Y p(X, Y)$ where $p(X)$ is the marginal probability of X
- ▶ **product rule:** $p(X, Y) = P(Y|X)p(X)$ where $P(X, Y)$ is the joint probability and $p(Y|X)$ is the conditional probability of Y given X
- ▶ **independence:** $p(X, Y) = p(X)p(Y)$ if X is independent from Y
- ▶ **Bayes' theorem:** $p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)}$

Quick recap on probabilities definitions

Bayes' theorem for model with parameters w :

$$p(w|\mathcal{D}) = \frac{p(\mathcal{D}|w)p(w)}{p(\mathcal{D})}$$

where

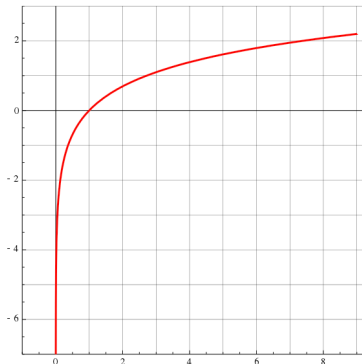
- ▶ w are the parameters of the model and $\mathcal{D}$ is the data
- ▶ $p(\mathcal{D}|w)$ is the likelihood that the model with parameters w has generated $\mathcal{D}$
- ▶ $p(w)$ is the prior probability of the model's parameters
- ▶ $p(w|\mathcal{D})$ is the posterior probability, i.e. the probability of the parameters after having observed $\mathcal{D}$
- ▶ $p(\mathcal{D})$ is the evidence

Quick recap on logarithm properties

The natural logarithm of x is the power to which e would have to be raised to equal x .

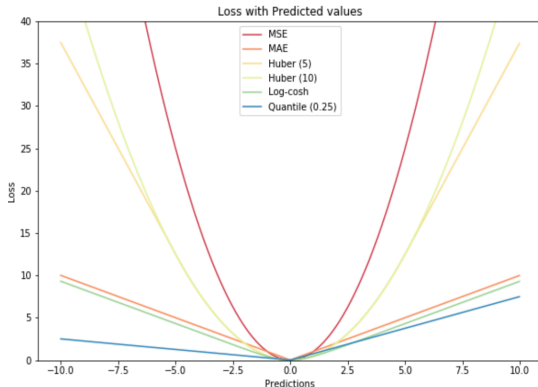
The following statements hold:

- ▶ $\ln e = 1$
- ▶ $\ln(uv) = \ln u + \ln v$
- ▶ $\ln \frac{u}{v} = \ln u - \ln v$
- ▶ $\ln u^n = n \ln u$
- ▶ $\ln e^u = u$



Error functions

- ▶ To improve the fit of a model we need to *measure* the quality of the fit w.r.t. some desired target via an error function
- ▶ An *error* function is a function that is 0 for a 0 input and positive elsewhere
- ▶ Generally error functions are differentiable, continuous, symmetric



In practical cases it might be better to use a function that has discontinuities

- ▶ A person who arrives before a plane gate closure can still board the plane, but a person who arrives after can not

or that are not symmetric

- ▶ In drug dosing, the cost of too little drug may be the complete lack of efficacy/protection, while the cost of too much may be some still tolerable toxicity

Error functions

The choice of the error function has consequences on the approximation that can be achieved/found by a learning machine

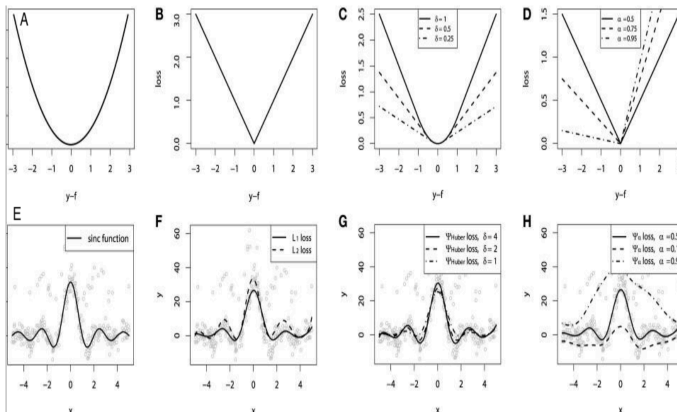


Figure: Natekin, Alexey, and Alois Knoll. "Gradient boosting machines, a tutorial." *Frontiers in neurorobotics* 7 (2013): 21

Q: How to choose among several options for error functions?

A: One criterion is to use the Maximum Likelihood principle

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Maximum Likelihood

Given some data, $\mathcal{D}$, and a model $p(x; w)$ of the data, what parameters w are most likely to have generated the data?

Likelihood Probability that a datum x was generated by p is given by the conditional probability

$$p(x | w)$$

Maximum Likelihood

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Likelihood Probability that a datum $\mathbf{x}$ was generated by p is given by the conditional probability

$$p(\mathbf{x} | \mathbf{w})$$

Assuming independence of observations, i.e. $\mathcal{D} = \{\mathbf{x}_n\}_{n=1}^N$ with independent $\mathbf{x}_n$, the overall likelihood for all the data is

$$\begin{aligned} p(\mathcal{D} | \mathbf{w}) &= p(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N | \mathbf{w}) \\ &= \prod_{n=1}^N p(\mathbf{x}_n | \mathbf{w}) \end{aligned}$$

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Maximum likelihood

Estimate $\mathbf{w}^*$ by finding the $\mathbf{w}$ that maximises $p(\mathcal{D} | \mathbf{w})$

Maximum Likelihood in the supervised setting

Maximise the likelihood that data $\mathcal{D} = \{(x_n, t_n)\}_{n=1}^N$ was generated by the model

$$\begin{aligned}\mathcal{L}(w) \equiv p(\mathcal{D} | w) &= \prod_n^N p(x_n, t_n | w) = \prod_n^N p(t_n | x_n, w) p(x_n | w) \\ &= \prod_n^N p(t_n | x_n, w) p(x_n)\end{aligned}$$

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Since log is monotonic, maximisation of $\mathcal{L}$ is equivalent to minimizing error function

$$E(w) = -\ln \mathcal{L}(w) = -\sum_n^N \ln p(t_n | x_n, w) - \sum_n^N \ln p(x_n)$$

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Since $p(x_n)$ does not depend on w , we can ignore it when minimising E

$$E(w) = -\sum_n^N \ln p(t_n | x_n, w)$$

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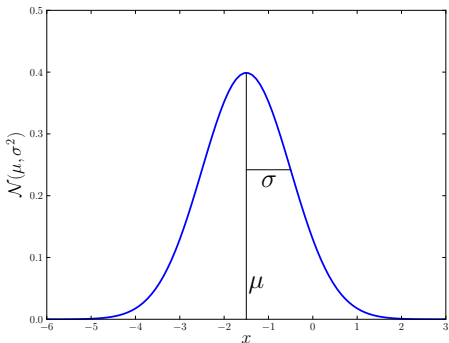
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Learning the mean of a normal density

Given data $\mathcal{D} = \{x_n\}_{n=1}^N$ estimate the mean.

Model

the data come from a Gaussian density of known variance σ^2

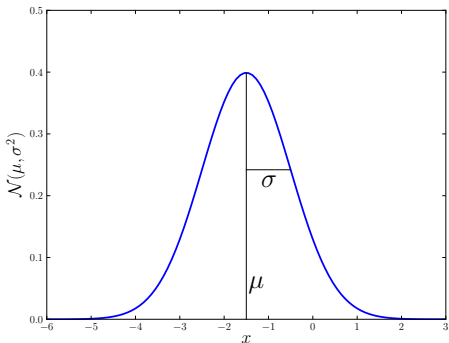


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Model likelihood:

$$p(x_n | \mu) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -(x_n - \mu)^2 / 2\sigma^2 \right\}$$

We wish to infer μ , assuming that σ is known.

$$p(x_n | \mu) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -(x_n - \mu)^2 / 2\sigma^2 \right\}$$

Likelihood for a set of N independent observations:

$$p(\mathcal{D} | \mu) = p(x_1, x_2, \dots, x_N | \mu) = \prod_{n=1}^N p(x_n | \mu)$$

Maximum likelihood

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Log likelihood

$$\begin{aligned} \ln \mathcal{L} = \ln p(\mathcal{D} | \mu) &= \sum_{n=1}^N \ln p(x_n | \mu) \\ &= -\frac{N}{2} \ln(2\pi\sigma^2) - \sum_{n=1}^N \frac{(x_n - \mu)^2}{2\sigma^2} \end{aligned}$$

Maximum likelihood

Maximise $\mathcal{L}$ or, equivalently, $\ln \mathcal{L}$:

$$\frac{d \ln \mathcal{L}}{d\mu} = \sum_{n=1}^N \frac{x_n - \mu}{\sigma^2} = 0$$

or

$$\sum_{n=1}^N x_n - \sum_{n=1}^N \mu = 0$$

So

$$\mu = \frac{1}{N} \sum_{n=1}^N x_n$$

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Alternative densities for $p(x)$ would yield different estimates for the mean.

Kullback-Leibler divergence

$$D(p, q) = \int p(x) \log \frac{p(x)}{q(x)} dx$$

- ▶ Expected value of p/q with respect to p
- ▶ Information gain about x that can be achieved by using $p(x)$ rather than $q(x)$
- ▶ $D(p, q) \neq D(q, p)$
- ▶ $D(p, q) \geq 0$
- ▶ $D(p, q) = 0$ iff $p(x) = q(x)$ almost everywhere
- ▶ Sometimes written $KL[p||q]$ or $D_{KL}(p||q)$ to emphasise it is a *divergence* between *densities* not a true distance.

KL divergence and maximum likelihood

- ▶ Suppose observations $\{x_n\}_{n=1}^N$ are sampled from density $p_0(x)$.
- ▶ Model data by a family of densities $p(x|w)$ parameterised by w
- ▶ All densities that can be modeled are $\mathcal{M} = \{p(x|w) : w \in W\}$
- ▶ If true density is contained in $\mathcal{M}$ then there is a parameter w_0 for which $p(x|w_0) = p_0(x)$

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- ▶ If true density is contained in $\mathcal{M}$ then there is a parameter w_0 for which $p(x|w_0) = p_0(x)$
- ▶ If $p_0(x) \notin \mathcal{M}$ then minimise KL divergence between true density and model:

$$\begin{aligned} KL[p_0||p] &= \int p_0(x) \log \frac{p_0(x)}{p(x)} dx \\ &= \int p_0(x) \log p_0(x) dx - \int p_0(x) \log p(x) dx \end{aligned}$$

KL divergence and maximum likelihood

$$KL[p_0||p] = \int p_0(x) \log p_0(x) dx - \int p_0(x) \log p(x) dx$$

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KL divergence and maximum likelihood

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- ▶ First term depends only on p_0
- ▶ Second term is estimated by the log likelihood for the data sample:

$$\int p_0(x) \log p(x) dx \approx \frac{1}{N} \sum \log p(x | w)$$

because integral is the expected value of $\log p(x | w)$ with respect to p_0 , which is estimated by the average because observations x_n are sampled from p_0 .

KL divergence and maximum likelihood

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- ▶ Maximising the (log) likelihood maximises the second term and therefore minimises $KL[p_0||p]$; **maximum likelihood approximates the true density as closely as possible within the family $\mathcal{M}$.**

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Regression example: Steam use

Operators of an industrial plant wish to predict the number y of pounds of steam used per month. It is believed that the quantity depends upon:

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$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

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The observed usage t_n month n in can be modelled as an observation from a Normal distribution with mean y_n and variance σ^2 .

$$\mathbf{x} = (x_1, x_2)$$

$$\mathbf{w} = (\beta_0, \beta_1, \beta_2, \beta_3, \sigma^2)$$

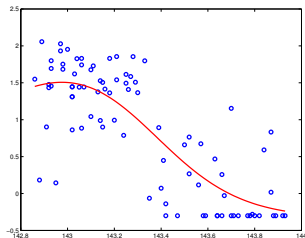
$$p(t_n | \mathbf{x}_n, \mathbf{w}) = (2\pi\sigma^2)^{-1/2} \exp \left\{ -\frac{[t_n - y(\mathbf{x}_n; \mathbf{w})]^2}{2\sigma^2} \right\}$$

Regression error function: accounting for noise

Model data generation as sum of deterministic function and Gaussian noise:

$$t_n = \mathcal{F}(x_n, w) + \epsilon_n \quad \epsilon_n \sim \mathcal{N}(0, \sigma^2)$$

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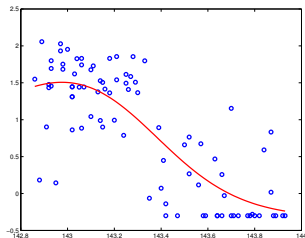
$$t_n = y_n + \epsilon_n$$

If $\mathcal{F}$ models the data *without* noise,

$$p(t_n | x_n, w) \propto \exp \left\{ -\frac{1}{2\sigma^2} (t_n - y_n(w))^2 \right\}$$

The error function can be expressed as a sum of squared errors

$$E(w) = - \sum_n \ln p(t_n | x_n, w) = \frac{1}{2\sigma^2} \sum_n (t_n - y_n(w))^2$$



- ▶ The residual sum of squares (RSS) is the sum of the squares of the residuals, that is of the deviations between the predicted vs. the actual empirical values of data
- ▶ RSS is a measure of the discrepancy between the data and an estimation model

$$\text{RSS} = \sum_{n=1}^N (t_n - f(x_n))^2$$

So, the minimisation of the RSS is equivalent to the maximisation of the likelihood of the model under the assumption of Gaussian noise.

Classification example: Heart problems

Health researchers wish to relate incidences of heart problems to:

- x_1 number of cigarettes smoked per day;
- x_2 number of units of alcohol consumed per day;
- x_3 an individual's age.

A model for the *systematic* component might be:

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Whether the individual n has a heart problem ($t_n = 1$) or not ($t_n = 0$) can be modelled as the outcome of the spin of a biased coin (a Bernoulli trial) with probabilities:

$$P(t_n = 1) = \exp(z_n) / [1 + \exp(z_n)]; \quad P(t_n = 0) = 1 / [1 + \exp(z_n)]$$

$$p(t_n | x_n, w) = \left(\frac{\exp(z_n)}{1 + \exp(z_n)} \right)^{t_n} \left(\frac{1}{1 + \exp(z_n)} \right)^{1-t_n}$$

Classification error function

Two classes, $\mathcal{C}_1$ and $\mathcal{C}_2$.

Classifier $y = \mathcal{F}(x, w)$ estimates posterior probabilities of class membership.

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Targets coded as

$$t = 1 \quad \text{if} \quad x \in \mathcal{C}_1$$

$$t = 0 \quad \text{if} \quad x \in \mathcal{C}_2$$

Posterior probabilities:

$$p(t = 1 | x) = p(\mathcal{C}_1 | x) = y$$

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$$p(t | x) = y^t (1 - y)^{1-t}$$

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For N independent observations, $\{t_n, x_n\}_{n=1}^N$:

$$p(T | X, w) = \prod_{n=1}^N y_n^{t_n} (1 - y_n)^{1-t_n}$$

Cross entropy error function

Define an error function as

$$E(w) = -\log p(T | X, w)$$

so that minimising $E(w)$ maximises the likelihood of the model.

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$$\begin{aligned} E(w) &= -\log p(T | X, w) \\ &= -\sum_{n=1}^N [t_n \log y_n + (1 - t_n) \log(1 - y_n)] \end{aligned}$$

Cross entropy error function

The cross-entropy of the distribution q relative to a distribution p over a given set is defined as follows:

$$H(p, q) = -E_p[\log q]$$

where $E_p[\cdot]$ is the expected value with respect to the distribution p . If we set up our notation, like $p \in \{t_n, 1 - t_n\}$ and $q \in \{y_n, 1 - y_n\}$ then the cross-entropy will give us a measure of dissimilarity between p and q :

$$H(p, q) = -\sum_i p_i \log q_i = -t_n \log y_n - (1 - t_n) \log(1 - y_n)$$

so that maximizing the likelihood is the same as minimizing the cross-entropy.

$$E(w) = -\log p(T | X, w) = \sum_{n=1}^N H(p, q)$$

Relationship between Cross entropy and KL

1. Cross-Entropy:

$$H(P, Q) = - \sum_{x \in \mathcal{X}} P(x) \log Q(x)$$

where $P(x)$ is the true probability distribution, $Q(x)$ is the predicted probability distribution, and $\mathcal{X}$ is the set of possible outcomes.

2. Kullback-Leibler Divergence:

$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}$$

where $P(x)$ is the true distribution and $Q(x)$ is the approximating distribution.

3. Entropy:

$$H(P) = - \sum_{x \in \mathcal{X}} P(x) \log P(x)$$

This represents the intrinsic uncertainty in the distribution $P(x)$.

Relationship between Cross entropy and KL

The cross-entropy can be expressed as the sum of the entropy of P and the KL divergence from P to Q :

$$H(P, Q) = H(P) + D_{\text{KL}}(P\|Q)$$

This decomposition explains their connection:

- ▶ **Entropy** $H(P)$: Represents the irreducible uncertainty inherent in P .
- ▶ **KL Divergence** $D_{\text{KL}}(P\|Q)$: Represents the extra cost (inefficiency) due to using Q instead of P .

Thus, the cross-entropy is always greater than or equal to the entropy of P , with equality only when $Q = P$ (i.e., $D_{\text{KL}}(P\|Q) = 0$).

When minimizing the cross-entropy loss in classification tasks, you're indirectly minimizing the KL divergence between the true label distribution P and the predicted distribution Q .

Modelling

- ▶ Systematic component – average response
- ▶ Random component – variability of observations

Maximum likelihood

- ▶ Find the parameters which are most likely to have generated the data

Error functions

- ▶ Maximum likelihood results in the error function

$$E(w) = -\log p(t | x, w)$$